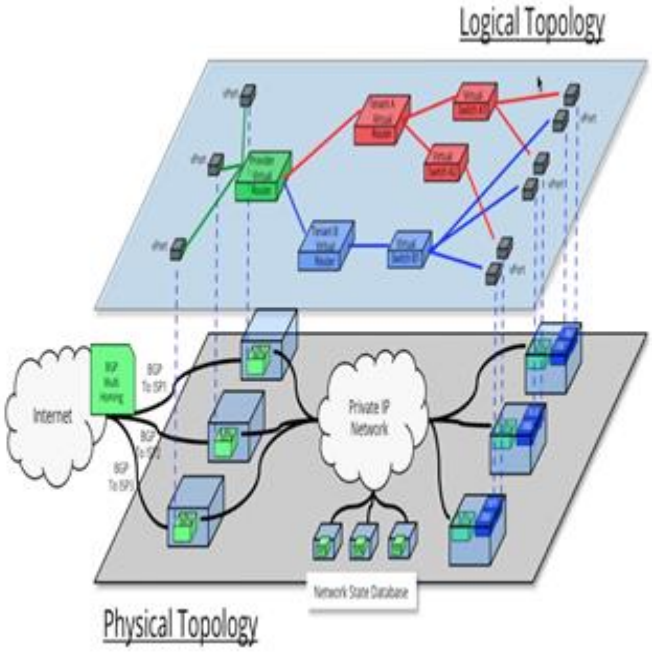


CLOUD COMPUTING COURSE

Lecture13-Network Virtualization in CC (2)

ABSTRACT

Lecture13 Introduces network devices and IP Address's

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1. IP Address and Mac Address

Definition of Mac Address

Address that uniquely defines a hardware interface is called **MAC (Media Access Control)** Address. MAC address is purchased by the manufacturer, producing interface hardware and assign the MAC addresses sequentially to the interface hardware as they are produced. MAC address is burned into the ROM of **Network Interface Card** (NIC). NIC is an interface hardware that is used by the computer to become a part of a network.

MAC address is a **48-bit hexadecimal** address. The format of a MAC address is MM:MM:MM:SS:SS:SS, where MM:MM:MM is a 3-byte address of the manufacturer. On the other hand, SS:SS:SS is a serial number of NIC card. MAC Address of each computer on a network is unique. When you change or replace the NIC card of your computer, your MAC address also gets changed.

MAC address is used at the data link layer of OSI/TCP/IP model. **ARP** (Address Resolution Protocol) is a protocol used to receive MAC address of a device.

Definition of IP Address

The address provided to a connection in a network is called **IP (Internet Protocol)** address. IP address does not uniquely identify a device on a network but, it specifies a particular connection in a network. IP address is provided by the administrator of the network or by Internet Service Provider (ISP).

IP address identifies both a network and the host on that network. IP address is used while routing as it specifically identifies a network connection. If your computer is on two networks so, it will have two IP addresses.

IPv4 address is 32-bit address whereas **IPv6** is 128-bit address. Your IP address will get changed each time you connect to the network as it is dynamically allocated to your device when it participates in the network. IP address for a particular connection in a network can be retrieved by RARP (Reverse Address Resolution Protocol).

Comparison Chart

BASIS FOR COMPARISON	MAC	IP
Full Form	Media Access Control Address.	Internet Protocol Address.
Purpose	It identifies the physical address of a computer on the internet.	It identifies connection of a computer on the internet.
Bits	It is 48 bits (6 bytes) hexadecimal address.	IPv4 is a 32-bit (4 bytes) address, and IPv6 is a 128-bits (16 bytes) address.
Address	MAC address is assigned by the manufacturer of NIC card.	IP address is assigned by the network administrator or Internet Service Provider.

Conclusion

MAC and IP address both are equally required when a device wants to communicate with another device in a network.

2. Physical Network Devices

Hardware devices that are used to connect computers, printers, fax machines and other electronic devices to a network are called **network devices**. These devices transfer data in a fast, secure and correct way over same or different networks. Network devices may be inter-network or intra-network. Some devices are installed on the device, like NIC card or RJ45 connector, whereas some are part of the network, like router, switch, etc. Let us explore some of these devices in greater detail.

1. Modem

Modem is a device that enables a computer to send or receive data over telephone or cable lines. The data stored on the computer is digital whereas a telephone line or cable wire can transmit only analog data.



Digital Data Waveform



Analog Data Waveform

The main function of the modem is to convert digital signal into analog and vice versa. Modem is a combination of two devices − **modulator** and **demodulator**. The **modulator** converts digital data into analog data when the data is being sent by the computer. The **demodulator** converts analog data signals into digital data when it is being received by the computer.

Types of Modem

Modem can be categorized in several ways like direction in which it can transmit data, type of connection to the transmission line, transmission mode, etc.

Depending on direction of data transmission, modem can be of these types −

- **Simplex** − A simplex modem can transfer data in only one direction, from digital device to network (modulator) or network to digital device (demodulator).
- **Half duplex** − A half-duplex modem has the capacity to transfer data in both the directions but only one at a time.
- **Full duplex** − A full duplex modem can transmit data in both the directions simultaneously.

2. RJ45 Connector

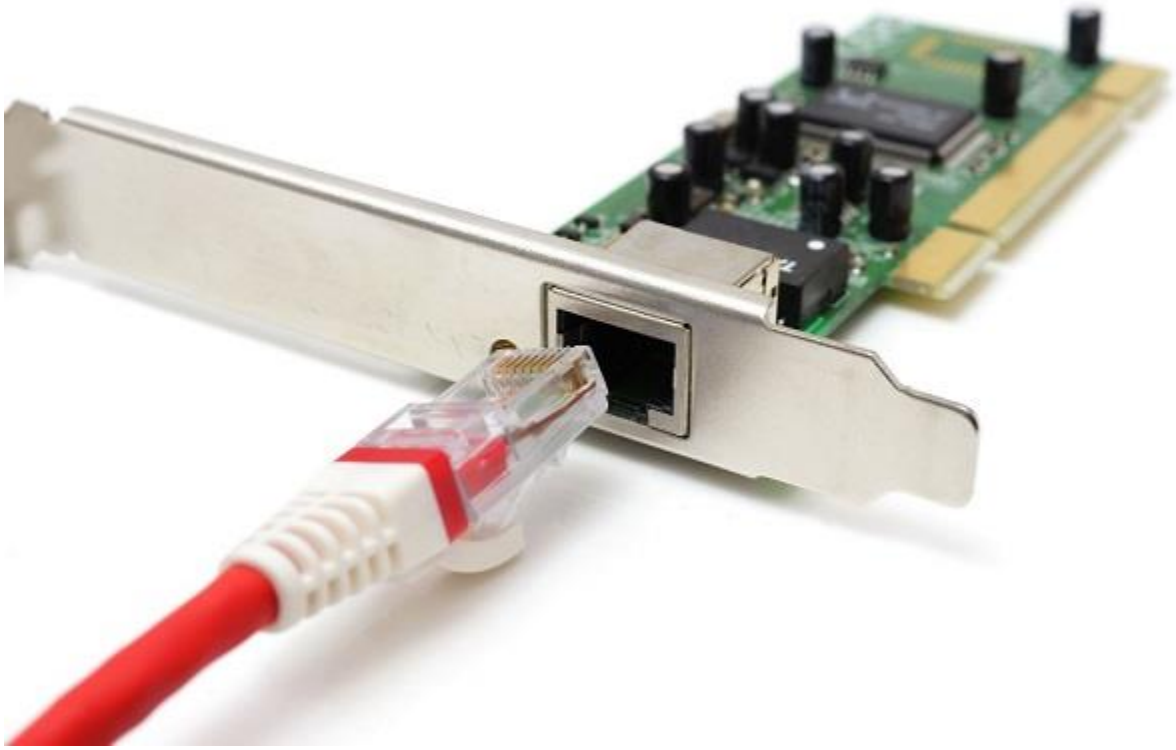
RJ45 is the acronym for **Registered Jack 45. RJ45 connector** is an 8-pin jack used by devices to physically connect to **Ethernet** based **local area networks (LANs)**. **Ethernet** is a technology that defines protocols for establishing a LAN. The cable used for Ethernet LANs are twisted pair ones and have **RJ45 connector pins** at both ends. These pins go into the corresponding socket on devices and connect the device to the network.



Ethernet Card

Ethernet card, also known as **network interface card (NIC)**, is a hardware component used by computers to connect to **Ethernet LAN** and communicate with other devices on the LAN. The earliest **Ethernet cards** were external to the

system and needed to be installed manually. In modern computer systems, it is an internal hardware component. The NIC has **RJ45 socket** where network cable is physically plugged in.



Ethernet card speeds may vary depending upon the protocols it supports. Old Ethernet cards had maximum speed of **10 Mbps**. However, modern cards support fast Ethernets up to a speed of **100 Mbps**.

Some cards even have capacity of **1 Gbps**.

3. Router

A **router** is a **network layer** hardware device that transmits data from one LAN to another if both networks support the same set of protocols. So a **router** is typically connected to at least two LANs and the **internet service provider** (ISP). It receives its data in the form of **packets**, which are **data frames** with their **destination address** added. Router also strengthens the signals before transmitting them. That is why it is also called **repeater**.



Routing Table

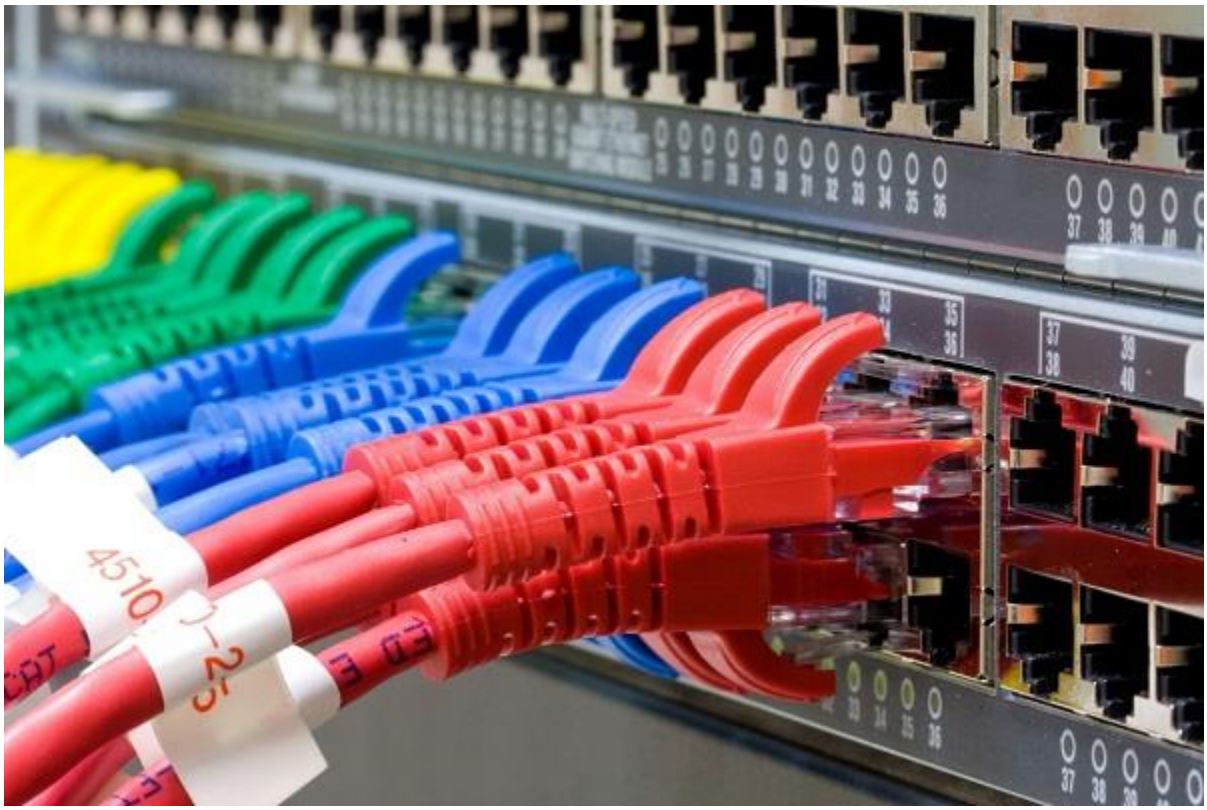
A router reads its routing table to decide the best available route the packet can take to reach its destination quickly and accurately. The routing table may be of these two types −

- **Static** − In a static routing table the routes are fed manually. So it is suitable only for very small networks that have maximum two to three routers.
- **Dynamic** − In a dynamic routing table, the router communicates with other routers through protocols to determine which routes are free. This is suited for larger networks where manual feeding may not be feasible due to large number of routers.

4. Switch

Switch is a network device that connects other devices to **Ethernet** networks through **twisted pair** cables. It uses **packet switching** technique to **receive, store** and **forward data packets** on the network. The switch maintains a list of network addresses of all the devices connected to it.

On receiving a packet, it checks the destination address and transmits the packet to the correct port. Before forwarding, the packets are checked for collision and other network errors. The data is transmitted in full duplex mode



Data transmission speed in switches can be double that of other network devices like hubs used for networking. This is because switch shares its maximum speed with all the devices connected to it. This helps in maintaining network speed even during high traffic. In fact, higher data speeds are achieved on networks through use of multiple switches.

5. Gateway

Gateway is a network device used to connect two or more dissimilar networks. In networking parlance, networks that use different protocols are **dissimilar networks**. A gateway usually is a computer with multiple **NICs** connected to different networks. A gateway can also be configured completely using software. As networks connect to a different network through gateways, these gateways are usually hosts or end points of the network.



Gateway uses **packet switching** technique to transmit data from one network to another. In this way it is similar to a **router**, the only difference being router can transmit data only over networks that use same protocols.

6. Wi-Fi Card

Wi-Fi is the acronym for **wireless fidelity. Wi-Fi technology** is used to achieve **wireless connection** to any network. **Wi-Fi card** is a **card** used to connect any device to the local network wirelessly. The physical area of the network which provides internet access through Wi-Fi is called **Wi-Fi hotspot**. Hotspots can be set up at home, office or any public space. Hotspots themselves are connected to the network through wires.



A **Wi-Fi card** is used to add capabilities like **teleconferencing, downloading** digital camera images, **video chat**, etc. to old devices. Modern devices come with their in-built **wireless network adapter**.

Related Video: *Please watch the next videos, for more explanation.*

1. ***IP Addresses***

https://youtu.be/ThdO9beHhpA

2. ***Network Devices***

https://youtu.be/1z0ULvg_pW8